Name:- Valle Ramesh

Case Study 2: Hospital Patient Care Analytics Pipeline

Scenario

A multi-specialty hospital wants to improve patient care, reduce waiting times, and predict high-risk patients. The hospital collects data from multiple systems such as patient registration, appointment scheduling, laboratory reports, wearable health devices, and doctor consultation notes.

The management wants to build a data-driven solution that can collect, process, store, and analyze patient data to improve operational efficiency and patient outcomes.

Your Task

Using the concepts learned in Data Engineering Fundamentals, design an end-to-end data engineering solution for the hospital.

Solution:-

(1.) Data Sources:- (Structured / Semi structured / Unstructured)

- **(i.)** Patient registration - Structured
- **(ii.)** Appointment Scheduling - Structured
- **(iii.)** Laboratory reports - Structured
- **(iv.)** wearable health devices - Structured
- **(v.)** doctor consultation notes - Unstructured

(2.) Data Ingestion:- (Batch / Stream / File - based)

(i.) Patient registration - Stream processing

Reason:- Registration details need to be evaluated immediately after registration.

(ii.) Appointment Scheduling - Stream processing

Reason:- Real time monitering needed for scheduling.

(iii.) Laboratory reports - File - based processing

Reason:- It includes scanning reports which is unstructured.

(iv.) wearable health devices - Batch processing

Reason:- Weekly (or) monthly records updated as a batch.

(v.) doctor consultation notes - Stream processing

Reason:- Consultation notes should update immediately.

(3.) ETL Process:-

(I.) Extract:- Data collected from

- **(i.)** Patient registration forms
- **(ii.)** Appointment Scheduling
- **(iii.)** Laboratory reports
- **(iv.)** wearable health devices
- **(v.)** doctor consultation notes

(II.) Transform:-

- **(i.)** Reschedule appointments which are at same time, based on their sevearity.
- **(ii.)** Remove duplicate registration forms.
- **(iii.)** Remove (or) replace invalid records.

(III.) Load:-

- **(i.)** Data Lake (Structured data)
- **(ii.)** Data Warehouse (Structured and processed data)

(4.) Data storage :-

(I.) Database:-

- Patients Information
- Appointment scheduling

(II.) Data Lake:-

- Labaratory reports
- Wearable health device records
- Medicine images

(III.) Data Data Warehouse:-

- Doctor consultation notes
- Weekly (or) Monthly recovery reports

(5.) Data Lakes :-

(i.) Raw Zone:- Registrations, Patient details, Laboratory reports

(ii.) Cleansed Zone:- Valid Registrations, Unique patient details, Lab reports with prescriptions.

(iii.) Curated Zone:- Analytics ready data.

(iv.) Sandbox Zone:- Performing operations using analytics ready data.

(6.**) Data Warehouse**:-

(i.) Business Reports Generated:-

- Patient Admitted Reports
- Appointment & Waiting Time Reports
- Laboratory Test Reports
- Doctor Performance Reports, Patient Treatment Reports
- Hospital Resource Utilization Reports

(ii.) <u>Departments Benefiting</u> : –

- **Doctors** – Patient history and risk analysis
- **Nursing Team** – Patient monitoring
- **Laboratory Department** – Test analysis
- **Emergency Department** – High-risk patient identification
- **Finance Department** – Treatment and billing analysis
- **Management** – Overall hospital performance

(7.) <u>Data Modeling and Data Pipelines</u>:-

(I.) <u>Patient Table</u>:- Patient ID (Primary Key) Name, Date of Birth, Gender, Phone, Address

(II.) <u>Appointment Table</u>:- Appointment ID **(Primary Key),** Patient ID **(Foreign Key)**, Doctor ID, Appointment Date, Appointment Status, Waiting Time.

(III.) <u>Laboratory Table</u>:- Lab Report ID **(Primary Key)**, Patient ID **(Foreign Key)**, Test Name, Test Result, Test Date.

<u>Relationship</u>:-

- **One Patient → Many Appointments**
- **One Patient → Many Lab Reports**

(8.) <u>Data Pipelines</u>:-

Patient Registration
↓
Appointment System
↓
Laboratory System
↓
Wearable Devices
↓
Doctor Consultation Notes
↓
Data Ingestion
↓
ETL

↓

Data Validation

↓

Data Lake

↓

Data Transformation

↓

Data Warehouse

↓

Analytics / Dashboards

↓

AI Models

↓

Business Decisions

(9.) <u>Data Quality and Validation</u>:-

(i.) Along with patient, second mobile number should be provided in the form.

(ii.) Each patient should have Unique ID on their report.

(iii.) Appointment date must be valid

(iv.) Lab results must have a valid patient's unique ID

(v.) Test results must follow the correct format

(vi.) Mandatory fields cannot be NULL

(10.) <u>Analytics Dashboards</u>:-

i. Patient Care Dashboard
ii. Waiting Time Dashboard
iii. Laboratory Dashboard
iv. Doctor Performance Dashboard

<u>AI Aplications</u>:- (I.) High-Risk Patient Prediction

(II.) Disease Risk Prediction

(III.) Patient Monitoring